BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: MRD-11TH 120623 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** MRD-11TH 120623 (A)
VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. Tan2α = ...

(A) $2\tan\alpha/(1-\tan^2\alpha)$
(B) $(1-\tan^2\alpha)/2\tan\alpha$
(C) $2\tan\alpha/(1+\tan^2\alpha)$
(D) $(1+\tan^2\alpha)/2\tan\alpha$

Q2. Sin(π/2 − α) = ...

(A) Sinα
(B) Cosα
(C) −Sinα
(D) Cosα

Q3. A triangle which has no right angle is called .....

(A) Oblique triangle
(B) Right angled triangle
(C) Both A and B
(D) None of these

Q4. $\sqrt{(s(s-a)/bc)}$ = ...

(A) Sinα/2
(B) Cosβ/2
(C) Cosα/2
(D) tanα/2

Q5. Period of tan3x is .....

(A) π
(B) 2π/3
(C) 2π
(D) π/3

Q6. $^{20}C_{18}$ = ...

(A) 360
(B) 190
(C) 40
(D) 10

Q7. 1 + oi is called .....

(A) Multiplicative identity
(B) Multiplicative inverse
(C) Additive inverse
(D) Additive identity

Q8. If z = x + iy and (z+5i)/(z−5i) = 1 then z lies on:

(A) x-axis
(B) y-axis
(C) Line y=5
(D) None of these

Q9. A system of linear equation has a unique solution, then the system is .....

(A) Inconsistent
(B) Consistent
(C) Both A and B
(D) None of these

Q10. If A and B are non-singular matrices. Then $(AB)^{-1}$ = ...

(A) $A^{-1}B^{-1}$
(B) $B^{-1}A^{-1}$
(C) $(BA)^{-1}$
(D) All of these

Q11. $\vec{A}$ = 2i + j − k then $|\vec{A}|$ = ...

(A) 2
(B) 4
(C) 6
(D) √6

Q12. The general term of a geometric sequence is .....

(A) a_1r^n
(B) a_1r^{n-1}
(C) a_1r^{n+1}
(D) $a_1+(n-1)d$

Q13. Relation between geometric and Harmonic mean is:

(A) G > H
(B) G < H
(C) G = H
(D) G = AH

Q14. What is the nth term of the series 1+(1+2)+(1+2+3)+.....?

(A) (n+1)/2
(B) n(n+1)/2
(C) $n^2(n^2+1)$
(D) (n+1)(2n+3)/2

Q15. If A = [1 3 1; −1 2 0; 2 0 −2] then A_{23} = ...

(A) 0
(B) −6
(C) 6
(D) 1

Q16. If $t_n = 6n+5$ then t_{n+1} = ...

(A) 6n−1 (B) 6n+11

(C) 6n+6 (D) 6n−5

Q17. (n+1)!/(n−1)! = ...

(A) n^2+n (B) (n+1)(n+2)

(C) 1/(n(n+1)) (D) (n+2)!/n

Q18. For any natural number n, 1 + 2 + 3 + 4 + ... + n = ...

(A) $\{n(n+1)/2\}^2$ (B) n(n+1)/2

(C) n(n+1)(n+2)/2 (D) n^2

Q19. The domain of $y = x/\sqrt{(x^2-3x+2)}$ is .....

(A) (∞,1) (B) (2,∞)

(C) (∞,1)U(2,∞) (D) (−∞,1)U(2,∞)

Q20. Which of the following point is a solution of inequality 2x−y≥ 0

(A) (−1,0) (B) (−1,−1)

(C) (1,−1) (D) All of these

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 50

Q2. Attempt any ten (10) parts each part carries equal marks:-

(i) Separate into real and imaginary parts $((4-5i)/(2-3i))^{-2}$.

(ii) Find the product of 2 − 3i and 7 + 5i.

(iii) Prove that $|1\ a\ a^2;\ 1\ b\ b^2;\ 1\ c\ c^2| = (a-b)(b-c)(c-a)$.

(iv) If A = [0 2 2; −1 3 2; 1 0 5], find A^{-1}.

(v) Find the angle between the vectors $\vec{OP}$ and $\vec{OQ}$ where $\vec{OP} = 2i + j$ and $\vec{OQ} = -3i + 2j$.

(vi) Find the volume of the parallelepiped whose edges are represented by a = 3i + j − k, b = 2i − 3j + k, c = i − 3j − 4k.

(vii) Find the sum of the first 200 positive odd integers.

(viii) Convert 0.123 (bar on 23) to a common fraction.

(ix) Find the sum to 2n terms of the series whose nth term is $4n^2 + 5n + 1$.

(x) Find n and r if ${}^nP_r = 840$ and ${}^nC_r = 35$.

(xi) Find domain and range of the function f(x) = (x−4)/(x−3).

(xii) Maximize $Z = 4x + 3y$ subject to the constraints $3x + 4y \leq 24$, $8x + 6y \leq 48$, $x \leq 5$, $y \leq 6$, $x,y \geq 0$

(xiii) Solve the equation $\tan\theta = -\sqrt{3}/3$

Section – C

Maximum Marks: 30

Note: Attempt any three (3) questions. Each question carries equal marks:-

Q3.

(a) Prove that $\text{Sin}2\theta = 2\tan\theta/(1+\tan^2\theta)$.

(b) Find the middle term in the expansion of $(2x^3 + 3y)^8$.

Q4.

(a) Solve the triangle with dimensions $\alpha = 100^o$, C = 345, $\gamma = 56.4^o$.

(b) Prove that $\text{Sin } 20^o \text{ Sin } 60^o \text{ Sin}40^o \text{ Sin } 80^o = 3/16$.

Q5.

(a) Three unbiased coins are tossed. What is the probability of obtaining at least one head?

(b) Without graphing, find vertex, all intercepts and axis of the graph of the function $f(x) = -x^2 + 6x + 5$.

Q6.

(a) Prove that $\text{Cos}(\tan^{-1}x) = 1/\sqrt{(1+x^2)}$, $x \geq 0$

(b) Find the unit vector perpendicular to both $i - 2j + 3k$ and $2i + j - k$.